MILITARY SPECIFICATION

BOOSTER, T23E1, LOADING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of booster, designated as Booster, T23E1, for use in high explosive ammunition.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

Military
MIL-B-60563 - Booster T23E1, Assembly Less Booster Pellet and Lead Cup Charge

STANDARDS

Military
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions

DRAWINGS
IEL-10523156 - Inspection Equipment List for Booster, T23E1
F10523156 - Booster, T23E1, Assembly
F7548175 - Box, Packing Ammunition for Booster Assembly, Details and Packing

(Copies of specification, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced standards and specifications.

3.2 First article sample.- (Not applicable).

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3.3 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specification and the following:

3.3.1 Foreign matter.- No part or assembly shall contain chips, dirt, grease, rust, corrosion, or other foreign matter.

3.3.2 Burr.- No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.2 First article sample.- (Not applicable).

4.3 Inspection provisions.

4.3.1 Submission of product.- The product shall be presented in accordance with MIL-STD-105.

4.3.2 Examination.- One-hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with MIL-STD-105. For classification of defects contained herein, the criteria in Table I shall be used. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burrs and foreign matter.

TABLE I

Defect Classification	Acceptable Quality Level (AQL)
Major (Booster)	AQL - 0.25 percent defective
Major (Packing and Marking)	AQL - 0.40 percent defective
Minor (Packing and Marking)	AQL - 0.65 percent defective

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4.3.2.1 Booster subassembly F10523003, covering a detail of Drawing F10523156.

Defect	Method of Inspection
CRITICAL	
1. Booster armed or partially armed	Visual
MAJOR	
101. Insecure, improper or incomplete assembly	Manual and Visual
102. Moisture in flash hole	Visual
103. Foreign matter in flash-hole	Visual
104. Booster damaged or defective	Visual
MINOR	
None defined	

4.3.2.2 Closing cup assembly, Drawing B8595506, covering a detail of Drawing F10523156.

CRITICAL	
None defined	
MAJOR	
101. Outside diameter exceeds max.	Gage
102. Total length exceeds max.	Gage
103. Explosive above flush	Gage
104. Explosive below flush exceeds max.	Gage
MINOR	
None defined	

4.3.2.3 Booster assembly, Drawing F10523156 (prior to assembly of booster pellet and cup).

CRITICAL	
1. Explosive material on small external threads of body.	Visual
2. Explosive material on internal thread of booster cup.	Visual

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4.3.2.3 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Closing cup charge flush or below	Gage
102. Closing cup assembly missing	Visual
103. Improper or incomplete crimp	Visual
104. Insecure or improper assembly	Manual and Visual
105. Foreign matter	Visual
MINOR	
None defined	

4.3.2.4 Booster assembly F10523156.

CRITICAL	
None defined	
MAJOR	
101. Total length exceeds max.	Gage
102. Diameter of booster cup exceeds max.	Gage
103. Staking missing or inadequate	Visual
104. Booster cup loosens when a disassembly torque of 35 inch pounds is applied	Gage
105. Booster cup cracked or split	Visual
106. Booster cup Pellet missing (an indication that booster pellet might be missing, if booster cup bottoms on shoulder of body assembly)	Visual
107. Booster pellet moves when assembly is shaken	Manual
108. Threads damaged or not full	Visual
109. Foreign matter	Visual
110. Marking missing, misleading or illegible	Visual
MINOR	
None defined	

4.3.2.5 Box, Set-up, Packing Ammunition for Boosters, Drawing C7548174, covering a detail of Drawing C7548175.

CRITICAL	
None defined	

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4.3.2.5 (Cont'd)

	Defect	Method of Inspection
MAJOR		
101.	Box damaged so that contents are exposed or liable to become exposed	Visual
102.	Sealing strip torn, badly wrinkled or incomplete	Visual
MINOR		
201.	Contents moves when shaken	Manual
202.	Marking missing, misleading, incorrect or illegible	Visual

4.3.2.6 Box, Packing Ammunition for Boosters, Drawing C7548175.

CRITICAL		
	None defined	
MAJOR		
101.	Board broken	Visual
102.	Hardware missing, broken or loose	Visual and Manual
103.	Contents exposed	Visual
104.	F.S.N. and D.O.D. identification misleading, incorrect or illegible	Visual
105.	Strapping missing, broken or loose	Visual and Manual
MINOR		
201.	Hardware improperly applied or engaged	Visual and Manual
202.	Contents move when shaken	Manual
203.	Strapping improperly applied	Visual and Manual
204.	Seal missing, unsealed or improperly positioned	Visual and Manual
205.	Marking misleading, incorrect or illegible	Visual

4.3.2.7 Process control audit.- Periodic unscheduled monitoring shall be conducted for the process operations on the following assemblies to determine compliance with the requirements of the applicable drawings.

a. Booster, T23E1, subassembly F10523003.

(1) Removal of dust seal

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4.3.2.7 (Cont'd)

b. Booster, T23E1, Assembly F10523156.

(1) Insertion of closing cup assembly

(2) Assembly of booster pellet and cup

(3) Application of sealing compound to cup threads

(4) Staking operation (if applicable)

4.3.3 Inspection testing.- The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Binder - lubricant content of tetryl.- A minimum of one sample (8 ounces) to a maximum of five samples (40 ounces) shall be required from each lot to determine the content of binder-lubricant of the tetryl in each lot of booster. Failure of the sample to meet the requirements specified on the applicable drawings shall be cause for rejection of the lot.

4.3.3.2 Specific gravity of booster pellet.- A sample of ten (10) booster pellets shall be selected from each lot of booster pellets to determine the specific gravity. Failure of one or more booster pellets to comply with the requirements specified on the applicable drawing shall be cause for rejection of the lot.

4.3.3.3 Inspection equipment.- The examinations and tests shall be made using equipment listed in IEL-10523156. Equipment design shall be in accordance with the code designation on the applicable IEL.

4.4 Test methods and procedures.

4.4.1 Binder-lubricant content of tetryl.- A sample of approximately 8 ounces of tetryl shall be selected and determined as follows: An accurately weighted portion of five (5) grams of the sample shall be transferred to a tared medium porosity sintered glass filtering crucible. The sample shall be extracted with four 10-milliliter portions of acetone. Each portion shall be maintained in contact with the residue for approximately five (5) minutes before applying suction. Any lumps which may persist shall be broken up. The side of the crucible shall be washed down with each 10 milliliter portion of acetone to insure complete removal of the tetryl. The residue shall be aspirated until the odor of acetone is no longer detectable. The crucible and contents shall be placed in a drying oven at 100 degrees (^{o}) to 105° Centigrade (C) for 1/2 hour, cooled in a desiccator and weighed. The above extraction, drying and weighing procedure shall be repeated with individual 10 milliliter portions of acetone until the loss in weight on subsequent extractions totals less than 0.005 gram. The weight of the residue in the crucible shall be calculated to the percentage of total binder-lubricant in the sample to determine compliance with the requirements specified on the applicable drawing.

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4.4.2 Specific gravity of booster pellet.- The representative booster pellets shall be cleaned so as to leave no sharp corners and shall then be accurately weighed on a laboratory balance. To determine the displacement, an apparatus (A8648248) consisting of a glass beaker with ground top, a piece of flat glass of sufficient size to completely cover the top of the beaker, and a semihard rubber perforated plate, pierced at each corner with sharp pointed pins so that about 1/2 inch of pin protrudes from each side, shall be used. The beaker shall be filled with clean mercury, the rubber plate inserted, and the mercury leveled off by pressing down the flat glass until it is even with the top of the beaker. The sample booster pellets shall be placed beneath the prongs of the rubber plate and the mercury leveled as before. The excess mercury shall be caught in a clean bowl. The mercury displaced by the pellet shall be calculated as follows to determine compliance with the requirement specified on the applicable drawing:

$$\text{Specific Gravity} = \frac{\sigma W}{W_1}$$

where σ = density of mercury at test temperature, in grams/cubic centimeter.
W = weight of booster pellets, in grams.
W_1 = weight of displaced mercury, in grams.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, labeling, and marking shall be as specified on drawing C-7548175.

5.2 Data Cards.- Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the title, number and date of this specification.

Custodian:

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Preparing activity:

Army-MU

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